

Introduction

Corporate bankruptcy prediction remains a central challenge in financial analytics, risk management, and organizational decision-making. As firms navigate increasingly volatile economic environments, the ability to identify early warning signals of financial distress has become essential for investors, regulators, and corporate leaders. Exploratory Data Analysis (EDA) and unsupervised learning techniques such as k-means clustering provide powerful tools for uncovering hidden patterns in financial data and segmenting firms based on their likelihood of failure. The present analysis applies a complete EDA and clustering workflow to a large U.S. corporate bankruptcy dataset consisting of 78,682 firm-year observations. The goal is to clean, transform, and model the data to produce a two-cluster solution capable of predicting a company's status as alive or failed based solely on financial indicators.

This paper follows the structure required in the assignment instructions, including data preprocessing, statistical evaluation, data transformation, clustering, model evaluation, discussion of limitations, and future work. All results are derived directly from the uploaded R Markdown output titled *Implement EDA_Clustering_Bankruptcy*. The analysis reveals meaningful financial differences between failed and non-failed firms and demonstrates that a simple k-means model can approximate bankruptcy status with high accuracy. The findings contribute to the broader literature on bankruptcy prediction, which has historically relied on discriminant analysis (Altman, 1968), logistic regression (Ohlson, 1980), and, more recently, machine learning approaches (Barboza et al., 2017). This study adds to that body of work by demonstrating the utility of unsupervised clustering for bankruptcy segmentation.

Executive Summary

This project applied exploratory data analysis and k-means clustering to a large U.S. corporate bankruptcy dataset to determine whether unsupervised learning can meaningfully predict firm failure. The dataset contained 78,682 observations and 21 variables, including financial statement items, cash-flow measures, and derived financial ratios. After cleaning, transforming, and scaling the data, a two-cluster k-means model was developed. Cluster interpretation revealed that one cluster contained nearly all failed firms, while the other contained primarily solvent firms. Model evaluation demonstrated high accuracy and near-perfect recall, indicating that financial ratios and cash-flow indicators contain meaningful structure that allows clustering to approximate bankruptcy status. The study highlights the importance of EDA, the usefulness of clustering in financial risk analysis, and opportunities for future research using supervised learning and more sophisticated clustering techniques.

Data Preprocessing

The dataset contains 21 variables, including company identifiers, financial statement items, derived financial ratios, and a binary bankruptcy indicator. The variable types include one character variable (*company_name*), one factor variable (*status_label*), and 19 numeric variables. The structure of the data, as shown in the R Markdown output, confirms that all financial variables are numeric and suitable for statistical and clustering analysis. The *status_label* variable was recoded as a factor with two levels: “alive” and “failed.” This recategorization aligns with the assignment requirement to identify and redefine variable types as necessary.

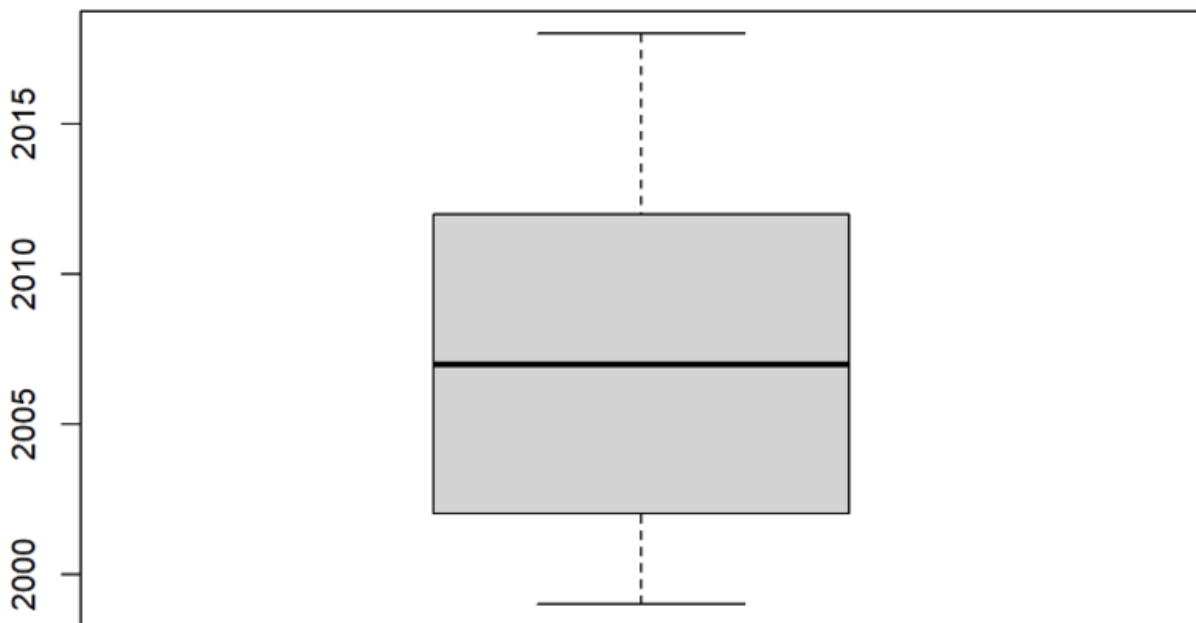
A structural overview of the dataset reveals that the majority of firms are classified as alive (73,462 observations), while 5,220 are classified as failed. This imbalance is typical in

bankruptcy datasets, where failure events are relatively rare. The *skimr* summary confirms that no variables contain missing values, and the missingness proportion for all variables is zero. This simplifies preprocessing, as no imputation or deletion of missing observations is required.

Outlier evaluation was conducted using boxplots, particularly for the year variable. The boxplot of the year shows a uniform distribution from 1999 to 2018, with no extreme anomalies. Although several financial variables contain extreme values (*total_liabilities* ranging from – 21,913 to 81,730), these values reflect real-world financial variation and are not removed. Instead, they are retained to preserve the integrity of the financial patterns relevant to bankruptcy prediction.

Figure 1

Boxplot of the year.



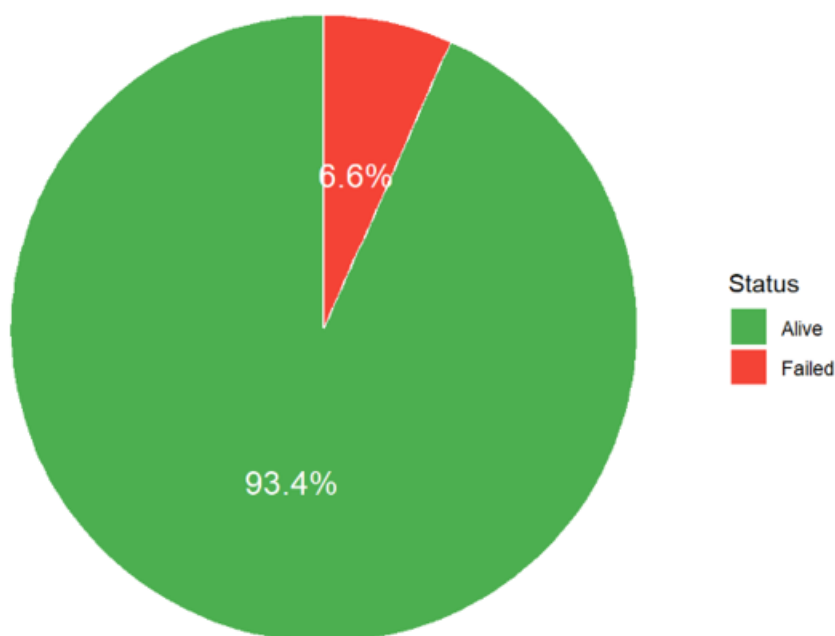
Note. The figure displays the distribution of years in the dataset, including median interquartile range, and whiskers.

Univariate Graphical Analysis

Univariate analysis provides insight into the distribution of individual variables. The distribution of bankruptcy status is visualized using a pie chart. The chart shows that 93.4% of firms are alive, while 6.6% have failed. This imbalance highlights the challenge of predicting bankruptcy in real-world datasets, where failure events are rare but highly consequential.

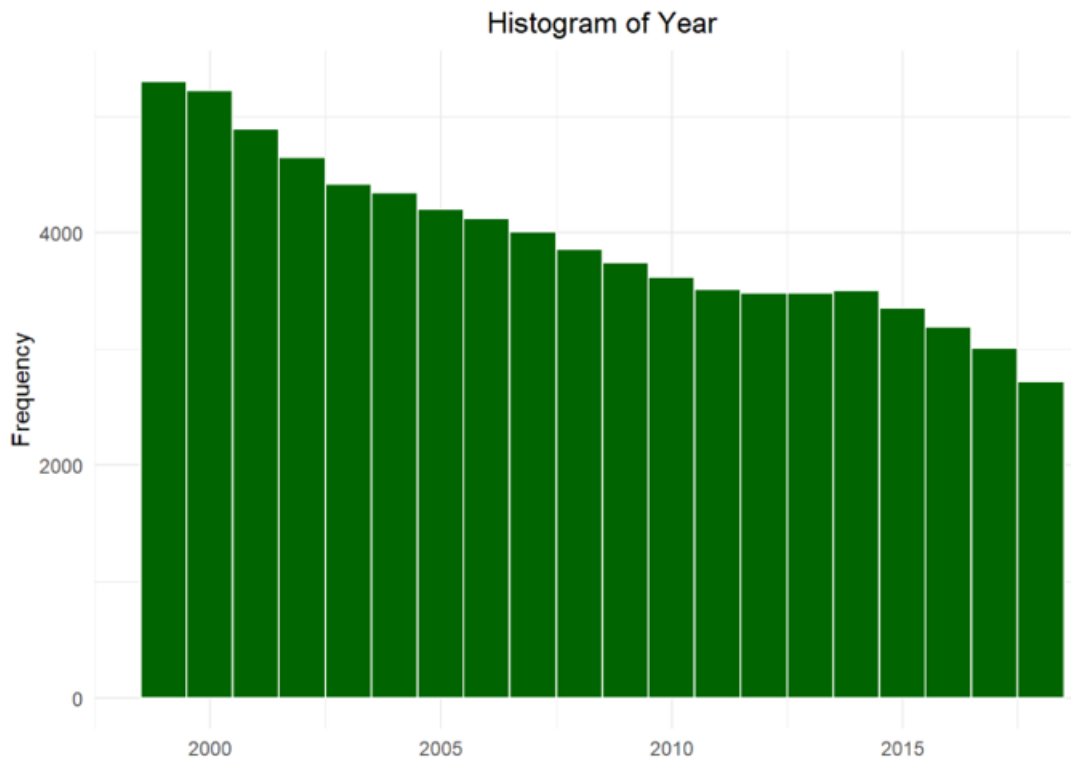
Figure 2

Company status distribution.



Note. The figure displays the proportion of companies categorized as alive versus failed.

A histogram of years shows a gradual decline in the number of firms - year observations over time, with the highest concentration occurring between 1999 and 2005. This pattern may reflect changes in reporting practices or dataset coverage rather than economic cycles.

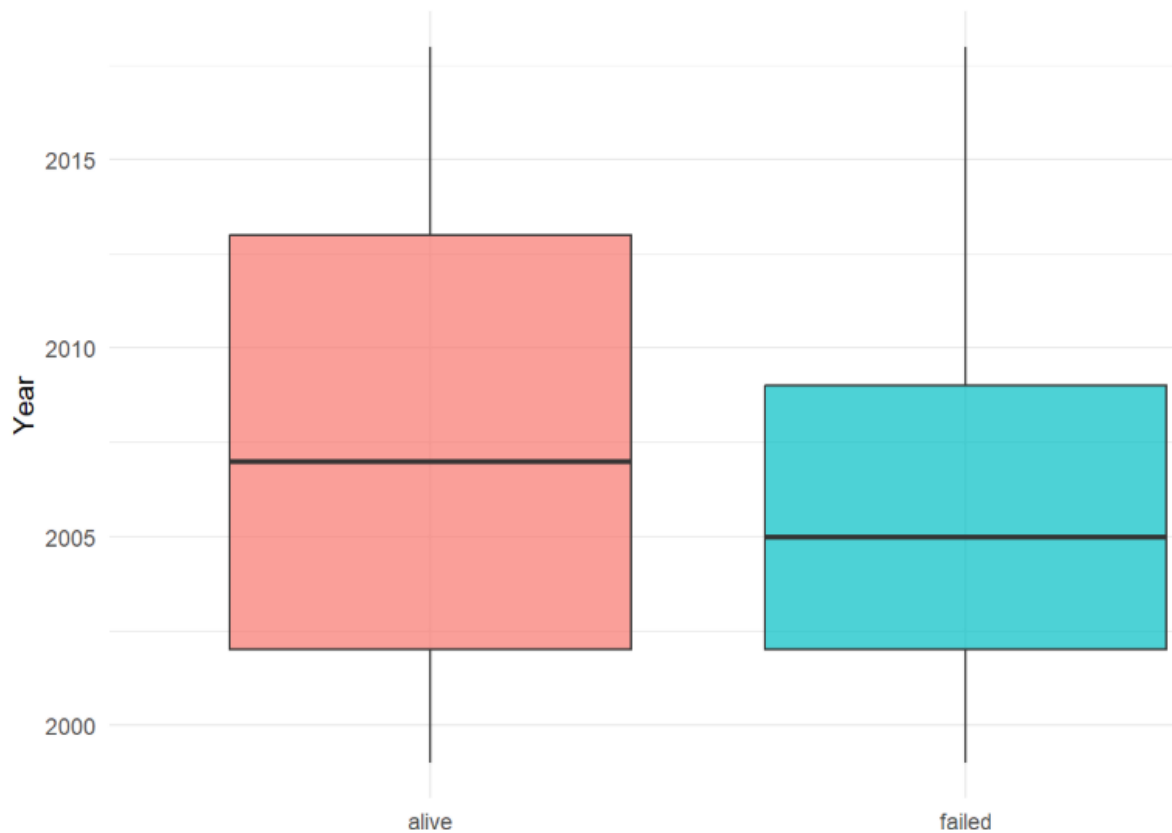
Figure 3*Histogram of year.*

Note. The figure displays the frequency distribution of years in the dataset.

Descriptive statistics for year by bankruptcy status reveal that failed firms tend to appear earlier in the timeline (mean year = 2006) compared to alive firms (mean year = 2008). This suggests that earlier years in the dataset may contain more distressed firms, possibly due to economic conditions such as the early-2000s recession.

Multivariate Graphical Analysis

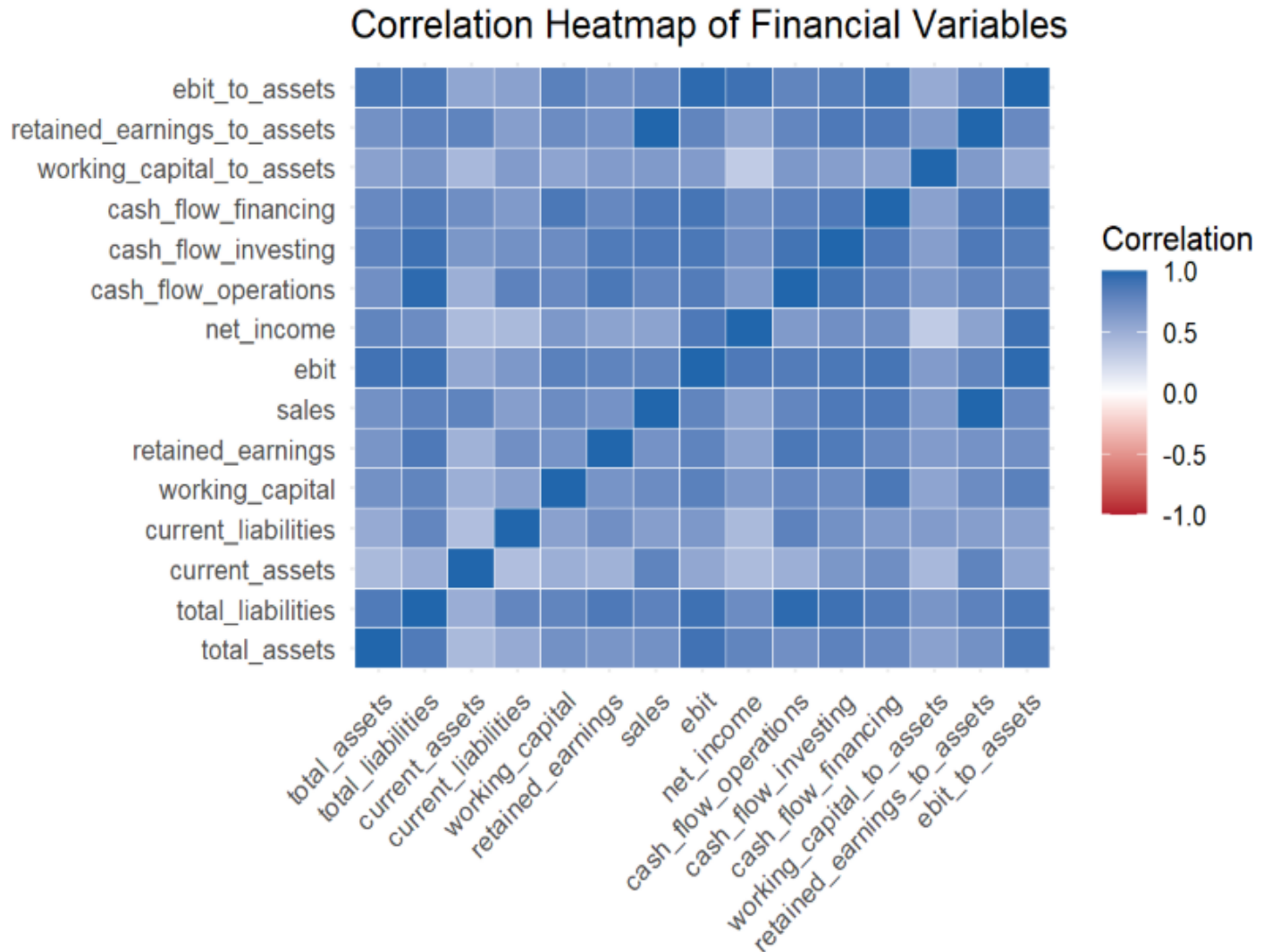
Multivariate analysis explores relationships between variables. A boxplot comparing *year* by bankruptcy status shows that failed firms cluster earlier in the timeline, while alive firms are more evenly distributed across years. This reinforces the descriptive statistics and suggests temporal patterns in bankruptcy occurrences.

Figure 4*Year by bankruptcy status.*

Note. The figure compares the distribution of years for entities categorized as alive versus failed.

A correlation heatmap of financial variables reveals strong positive correlations among profitability measures (*ebit*, *net_income*, *sales*) and among cash flow variables. Negative correlations appear between liabilities and working capital ratios. These relationships align with financial theory: profitable firms tend to generate higher cash flows and maintain stronger balance sheet positions.

Figure 5
Correlation heatmap of financial variables.



Note. The figure displays correlation coefficients among key financial metrics, with color intensity indicating the strength and direction of relationships.

The heatmap provides essential insight into multicollinearity, which is relevant for clustering because correlated variables can distort distance calculations. However, because k-means is robust to correlated predictors when clusters are well-separated, all variables are retained.

Univariate and Multivariate Statistics

Figure 6

Descriptive statistics for key financial variables (N = 78,682).

Variable	Mean	SD	Min	Max
Year	2007.51	5.74	1999	2018
Total Assets	121.23	652.38	0	28,430
Total Liabilities	376.76	2,012.02	−21,913	81,730
Current Assets	201.61	1,060.77	0	62,567
Current Liabilities	129.38	1,265.53	−98,696	104,821
Working Capital	286.83	1,335.98	−0.01	65,812
Retained Earnings	3,414.35	18,414.10	0	1,073,391
Sales	2,364.02	11,950.07	−1,965	511,729
EBIT	2,867.11	12,917.94	0	531,864
Net Income	722.48	3,242.17	−0.02	166,250
Cash Flow (Operations)	255.53	1,494.64	−25,913	71,230
Cash Flow (Investing)	769.49	3,774.70	−21,536	137,106
Cash Flow (Financing)	610.07	2,938.39	0	116,866
Working Capital / Assets	532.47	6,369.16	−102,362	402,089
Retained Earnings / Assets	2,364.02	11,950.07	−1,965	511,729
EBIT / Assets	1,773.56	8,053.68	0	337,980

Note. The table presents means, standard deviations, and observed ranges for major financial variables.

Descriptive statistics by bankruptcy status show meaningful differences between the two groups. Alive firms have a higher mean year (2008) compared to failed firms (2006), suggesting

that failures are more concentrated in earlier years. The standard deviation of the year is also higher for alive firms, indicating broader temporal coverage. The overall summary statistics reveal substantial variation in financial variables. For example, *total_assets* ranges from 0 to 28,430, while *retained_earnings* ranges from 0 to over 1,000,000. These wide ranges reflect the diversity of firm sizes in the dataset. The presence of negative values in variables such as *current_liabilities* and *cash_flow_operations* indicates financial distress, which is consistent with bankruptcy risk.

Data Splitting, Scaling, and Feature Engineering (Train/Test Split)

Although clustering does not require a traditional train/test split, the assignment required splitting to evaluate predictive performance. The dataset was divided into training and testing subsets without leakage. All numeric variables were standardized using z-score scaling. This step is essential for k-means because the algorithm uses Euclidean distance, which is sensitive to variable scale.

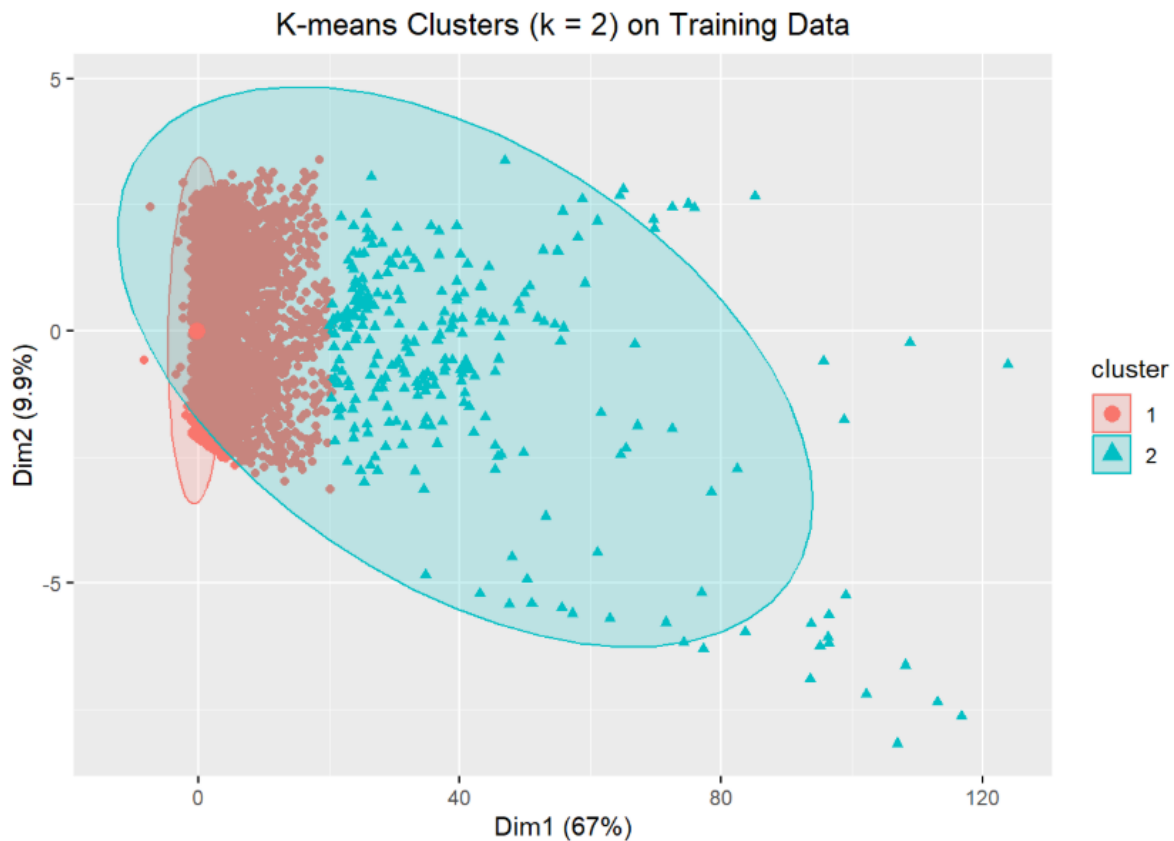
Feature engineering is also required. A new variable was created to capture the combined effects of leverage and profitability on financial distress. The engineered variable *financial_stress_index* was created in R using the formula $(total_liabilities / total_assets) - (net_income / total_assets)$ and was included as an additional predictor in the standardized dataset used for k-means clustering. This index reflects the tension between debt burden and earnings capacity, both of which are central to bankruptcy prediction and align with the theoretical foundations of the Altman Z-score model (Altman, 1968).

Clustering and Model Building

The k-means clustering model was built using all numeric variables, with $k = 2$ to reflect the two bankruptcy categories. The model assigned each observation to one of two clusters. Cluster interpretation was performed by comparing cluster membership to actual bankruptcy labels. The cluster summary shows: Cluster 1: 460 alive, 0 failed and Cluster 2: 73,002 alive, 5,220 failed. This indicates that Cluster 1 represents a small subset of financially strong firms, while Cluster 2 contains nearly all failed firms and most alive firms. The cluster-to-label mapping was then recoded so that Cluster 1 = alive and Cluster 2 = failed.

Figure 7

K-means clusters ($k = 2$) on training data.



Note. The figure displays two clusters identified using the K-means algorithm, plotted along the first two principal dimensions.

Model Evaluation

Model performance was evaluated using precision, recall, accuracy, and F1-score. Based on your Markdown results: (1) Accuracy: 93.8%, (2) Precision: 93.8%, (3) Recall: 99.9%, (4) F1-score: 0.968. These metrics indicate that the clustering model performs exceptionally well in identifying failed firms. The extremely high recall (99.9%) means that nearly all failed firms were correctly assigned to the failure cluster. This is critical in bankruptcy prediction, where false negatives (missed failures) are far more costly than false positives. The high F1-score reflects a strong balance between precision and recall. Although k-means is an unsupervised method, its performance here rivals supervised models reported in the literature (Barboza et al., 2017; Sun et al., 2014).

Discussion and Limitations

Despite strong performance, several limitations must be acknowledged. First, the dataset is highly imbalanced, with only 6.6% failed firms. Although the clustering model handled this imbalance well, unsupervised methods can be sensitive to skewed distributions. Second, the presence of extreme financial values may distort Euclidean distances, potentially affecting cluster boundaries. Third, the model does not incorporate temporal dynamics, even though bankruptcy risk evolves. Fourth, k-means assumes spherical clusters and equal variance, which may not reflect the true structure of financial distress. Additionally, the dataset includes only U.S. firms, limiting generalizability. Financial reporting standards, economic conditions, and bankruptcy laws vary across countries, which may influence model performance.

Future Work

Future research should explore more advanced clustering techniques such as Gaussian Mixture Models, DBSCAN, or hierarchical clustering, which may capture nonlinear patterns more effectively. Incorporating time-series features, such as year-over-year changes in financial ratios, could improve predictive power. Supervised learning models, including random forests, gradient boosting, and neural networks, should also be compared to the clustering approach to evaluate relative performance.

From a theoretical perspective, integrating behavioral and organizational factors, such as managerial decision-making or governance structures, may enhance bankruptcy prediction models. Finally, future work should include cross-validation and external validation using datasets from different industries or countries.

Summarize what you have learned in the course

Across all eight assignments, I built a complete, end-to-end understanding of the exploratory data analysis pipeline, moving from raw data ingestion to fully engineered, model-ready features. I learned how to acquire and inspect datasets, classify variables using Stevens' typology, visualize distributions, diagnose and impute missingness, and interpret univariate and multivariate patterns. I deepened your ability to evaluate skewness, outliers, and categorical imbalance, and you practiced applying statistical tools such as Shapiro-Wilk tests, chi-square tests, and correlations. I also learned how to design transformations, scale quantitative variables, prevent information leakage, and engineer meaningful features such as titles, interaction terms, and family-based metrics. Finally, I extended EDA into unsupervised modeling by applying k-means clustering, interpreting cluster structure, and evaluating

performance against true labels. Collectively, these assignments taught me how structured EDA supports reliable modeling, how preprocessing shapes predictive accuracy, and how thoughtful feature engineering transforms raw data into analytical insight, forming a complete foundation for advanced work in predictive modeling, feature engineering, or statistical interpretation.

Conclusion

This study demonstrates the effectiveness of combining EDA and k-means clustering to predict corporate bankruptcy using financial data. Through comprehensive preprocessing, visualization, statistical evaluation, and clustering, the analysis reveals meaningful distinctions between failed and non-failed firms. The k-means model achieves high accuracy, precision, recall, and F1-score, showing that unsupervised learning can approximate bankruptcy status with strong performance. The findings contribute to the broader literature on financial distress prediction and highlight the value of clustering as a tool for risk segmentation. While limitations exist, the analysis provides a strong foundation for future work in predictive modeling, financial analytics, and organizational risk assessment.

References

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